Two City Scheduling 2

A company is interviewing 2N candidates. Flying candidate i to city A costs costs[i][0] and to city B costs costs[i][1].

Exactly N people must go to city A and exactly N people must go to city B.

The candidates are processed in the exact order they appear in the input array. If two adjacent candidates in the input array are flown to the same city, an additional collaboration penalty of P is added to the total cost.

Find the minimum total cost to fly all candidates.

Note:

It is given that M is guaranteed to be even.

Input Format

The first line contains a integer, N, denoting the 2N, the total number of candidates.

The second line contains a integer, P, denoting the penalty for flying adjacent candidates to the same city.

Each of the N lines contains 2 space-separated integers, representing row i of costs.

Constraints

1 <= N <= 100

0 <= P <= 1000

1 <= costs[i][j] <= 1000

Output Format

The output contains an integer value

Sample Input 0

```
2
100
10 20
30 40
```

Sample Output 0

```
50
```

Explanation 0

The input has M = 2 candidates and a penalty P = 100. The costs for candidate 1 are 10 for city A and 20 for city B. The costs for candidate 2 are 30 for city A and 40 for city B. We must send exactly 1 candidate to each city. If we send the first to A and the second to B, the cost is 10 plus 40, which equals 50. Since they go to different cities, there is no penalty. Sending the first to B and the second to A costs 20 plus 30, which also equals 50 without penalty. The minimum total cost is 50.

Sample Input 1

```
4
50
10 100
10 100
100 10
100 10
```

Sample Output 1

```
140
```

Explanation 1

The input has M = 4 candidates and a penalty P = 50. The flight costs heavily favor sending the first two candidates to city A (costing 10 each instead of 100) and the last two to city B (costing 10 each instead of 100). If we follow this assignment, the base flight cost is 40. However, the first two candidates go to the same city, adding a penalty of 50. The last two also go to the same city, adding another penalty of 50. The total cost is 140. Alternatively, strictly alternating cities would cost 220 with no penalty. Thus, 140 is the minimum cost.

Sample Input 2

```
4
200
10 100
10 100
100 10
100 10
```

Sample Output 2

```
220
```

Explanation 2

The input uses the exact same candidate flight costs as the previous sample, but the penalty P is increased to 200. Sending the first two candidates to city A and the last two to city B yields a total cost of 440. If we instead assign them in the pattern A, B, A, B, the base flight costs become 220. Because no adjacent candidates go to the same city, the penalty is 0. The minimum total cost is therefore 220.